EXECUTION EVIDENCE · REAL ON-CHAIN TRANSACTION HASHES · ETHEREUM SEPOLIA

Shouhu 守护

Execution evidence: real on-chain transaction hashes on Ethereum Sepolia

2026 Shenzhen International FinTech Competition (FinTechathon) · International Track · Topic F: Agent Wallet — Autonomous On-chain Wallet

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Version 1.0 · 20 October 2026 · read from the chain on 2026-10-20 17:11 UTC+8 (Beijing time)

EVIDENCE AT A GLANCE

- **Network and contracts:** Ethereum Sepolia (chain id 11155111). PolicyWallet `0xb9D959da30680e86E9bF0A287Abee354033a0525` and TestUSDC `0x6b2f3F61DB4e45D07453D58dF2F4eB27C5284931`, both source-verified on Sourcify (exact match).
- **42 transactions:** 2 deployments, 11 rule changes, 2 deposits of test money (11,000 tUSDC), 6 one-time owner approvals, **20 payments by the AI agent (14 Tier 1 inside the limits, 6 Tier 2 with the parent's approval)**, and **3 transactions the contract rejected**.
- **Traceable:** 20 of 20 payments link to the student's words and the policy decision through the intent id written on-chain (audit log committed in the public repository).
- **Checked:** every wallet transaction passes the proof checks (37 of 37); no payment left the wallet outside policy; balance 5,980 tUSDC.
- **Machine-readable:** `transactions.csv` and `transactions.json` (every hash with its Etherscan link), `proof_checks.json`, `rejections.json`, `verify_all_payments.txt`, `audit_log.jsonl`.

Every payment the agent made is on-chain, inside the parent's policy, and traceable to the student's own words. The contract refused every attempt to go around the policy.

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tUSDC is a test token with no monetary value; all transactions are on the Ethereum Sepolia test network. Shouhu 守护 means "to guard".

1. How to check any hash

1. Open the hash on Etherscan (every hash in this report is a link). Check **Status: Success**, **From** (the agent key `0xe46A…b291` or the owner key `0x4444…9EEF`), **To** the PolicyWallet, and under **Logs** the `Payment` or `ApprovalGranted` event and the tUSDC `Transfer`.
2. Compare the contract source on Sourcify (PolicyWallet, exact match) with `contracts/src/PolicyWallet.sol` in the repository.
3. Read the intent id in the `Payment` event: it is the keccak-256 hash of the policy decision in the audit log (`evidence/runs/sepolia-20261020-124109-web-v2/audit_log.jsonl`). Recompute it, or run `uv run shouhu-verify --chain sepolia <hash>` to see the student's words, the decision, its tier and reasons.
4. Ask the live contract directly, with no keys: `uv run shouhu-try --chain sepolia mama 600` answers REJECTED (`ExceedsPerTransferCap`).

Repository: https://github.com/SeerieTheGreat/AIAgent. Contract addresses: PolicyWallet `0xb9D959da30680e86E9bF0A287Abee354033a0525`, TestUSDC `0x6b2f3F61DB4e45D07453D58dF2F4eB27C5284931`, owner key `0x4444b10a36E88682C63caE89E2ADb561651e9EEF`, agent key `0xe46Ab06dF31a8E03BAAe6aA5912b60B10e0Fb291`.

2. Deployment and setup

Time (UTC+8)	What happened	Signed by	Transaction
10-16 18:15	deploy TestUSDC (0x6b2f3F61…)	owner key (parent)	`0x09fc9fe7…fdaee2`
10-16 18:15	deploy PolicyWallet (0xb9D959da…)	owner key (parent)	`0xa858a9a4…f6d9b9`
10-16 18:15	AgentUpdated: agent key 0xe46Ab06dF31a8E03BAAe6aA5912b60B10e0Fb291	owner key (parent)	`0xa858a9a4…f6d9b9`
10-16 18:15	PolicyUpdated: limits 500 tUSDC per transfer, 1000 tUSDC per 24 hours	owner key (parent)	`0xa858a9a4…f6d9b9`
10-16 18:16	Transfer into the wallet (test money minted): 10000 tUSDC into the PolicyWallet	owner key (parent)	`0x14d03120…e45df2`
10-16 18:16	AllowlistUpdated: Mama added to the allowlist	owner key (parent)	`0xd1c15e59…1c43dc`
10-16 18:16	AllowlistUpdated: Ge added to the allowlist	owner key (parent)	`0xb66b24a2…5de605`
10-16 18:16	AllowlistUpdated: Landlord Zhang added to the allowlist	owner key (parent)	`0x9a886db8…6c35fb`
10-16 18:16	AllowlistUpdated: University Bursar added to the allowlist	owner key (parent)	`0xbce911f4…29483e`
10-16 18:17	AllowlistUpdated: Xiao Wang added to the allowlist	owner key (parent)	`0x19c6182f…63cd48`

The PolicyWallet constructor set the agent key and the limits (500 tUSDC per transfer, 1,000 tUSDC per 24 hours) in its deployment transaction; the owner then minted 10,000 tUSDC of test money into the wallet and allowlisted the five default contacts.

3. Payments by the AI agent

3.1 Tier 1: inside the parent's limits (14 payments)

Signed by the agent key with `agentTransfer`. The policy engine allowed each one, and the contract re-checked the allowlist, the per-transfer cap and the rolling 24-hour cap.

Time (UTC+8)	Payment	The student's words	Transaction
10-16 18:25	20 tUSDC to Mama	"Send 20 USDC to Mama."	`0xb9ee7295…520074`
10-16 18:27	10 tUSDC to Ge	"给哥转 10 USDC"	`0xbb784709…a5dc7c`
10-19 15:18	20 tUSDC to Mama	"Send 20 USDC to Mama."	`0x3330ce7a…45ca06`
10-19 15:20	10 tUSDC to Ge	"给哥转 10 USDC"	`0xe50ff1be…ed42aa`
10-20 01:37	20 tUSDC to Mama	"Send 20 USDC to Mama."	`0xe21b79ff…b5363f`
10-20 01:42	10 tUSDC to Ge	"给哥转 10 USDC"	`0x0352e2f7…3cc6c7`
10-20 06:51	15 tUSDC to Aunt Li 李阿姨	"给李阿姨转 15 USDC"	`0x4830a81f…c8c332`
10-20 07:00	20 tUSDC to Mama	"Send 20 USDC to Mama."	`0x9d9d76a5…247ffb`
10-20 07:02	15 tUSDC to Aunt Li 李阿姨	"给李阿姨转 15 USDC"	`0xb8a35a26…e23213`
10-20 07:05	20 tUSDC to Mama	"Send 20 USDC to Mama."	`0x2f126f72…af39ce`
10-20 07:10	15 tUSDC to Aunt Li 李阿姨	"给李阿姨转 15 USDC"	`0x53f802b5…d3c0a3`
10-20 12:19	20 tUSDC to Mama	"Send 20 USDC to Mama."	`0xc4007a1c…c51859`
10-20 12:20	10 tUSDC to Ge	"给哥转 10 USDC"	`0xe4e2a1f3…79a81b`
10-20 12:28	15 tUSDC to Aunt Li 李阿姨	"给李阿姨转 15 USDC"	`0x62e6726e…45ac3f`

3.2 Tier 2: with the parent's one-time approval (6 payments)

Each request went above the 500 tUSDC per-transfer cap, so the policy engine filed it for the parent. The parent approved exactly that recipient and amount once with the owner key (`approveOnce`), and the agent then executed it with `agentTransferWithApproval`, which marks the approval used.

Time (UTC+8)	Payment	The student's words → reason for approval	Parent's approval	Payment
10-16 18:26	800 tUSDC to Landlord Zhang	"Pay the landlord 800 USDC for the apartment deposit." → 800 tUSDC exceeds the per-transfer cap of 500 tUSDC	`0xd4961bcc…4b0509`	`0x838aee65…a38cfc`
10-19 15:19	800 tUSDC to Landlord Zhang	"Pay the landlord 800 USDC for the apartment deposit." → 800 tUSDC exceeds the per-transfer cap of 500 tUSDC	`0x16f6267e…dc96d1`	`0x4d6b84ae…87ab48`
10-20 01:39	800 tUSDC to Landlord Zhang	"Pay the landlord 800 USDC for the apartment deposit." → 800 tUSDC exceeds the per-transfer cap of 500 tUSDC	`0x18fa60fe…aafaa2`	`0xf729b1d8…f5aa68`
10-20 07:01	800 tUSDC to Landlord Zhang	"Pay the landlord 800 USDC for the apartment deposit." → 800 tUSDC exceeds the per-transfer cap of 500 tUSDC	`0x4a0f6659…c18c27`	`0xc4805f36…e8b8ba`

Time (UTC+8)	Payment	The student's words → reason for approval	Parent's approval	Payment
10-20 07:08	800 tUSDC to Landlord Zhang	"Pay the landlord 800 USDC for the apartment deposit." → 800 tUSDC exceeds the per-transfer cap of 500 tUSDC	`0x8d783a7b…d636a7`	`0xdec61abe…ab1377`
10-20 12:23	800 tUSDC to Landlord Zhang	"Pay the landlord 800 USDC for the apartment deposit." → 800 tUSDC exceeds the per-transfer cap of 500 tUSDC	`0x499b22f3…81e58c`	`0x8c592d12…b9d7fe`

The approval and the payment share one intent id, and the same id identifies the policy decision in the audit log. A second attempt with the same approval reverts with `ApprovalUsed` (contract test `test_approvalFlow`).

4. Transactions the contract rejected

To show what happens if the software above the contract is bypassed or compromised, three transactions were sent directly with the agent key on 20 October 2026, skipping the policy engine and the app's dry run. The contract rejected all three; nothing moved, and the wallet balance stayed at 5,980 tUSDC.

Attempt by the agent key	Rejected with	Status	Transaction
600 tUSDC to Mama (per-transfer cap is 500)	`ExceedsPerTransferCap(amount=600000000, cap=500000000)`	Failed (reverted)	`0xd1a63517…4c2319`
1 tUSDC to the scam address from attack A1 (not allowlisted)	`NotAllowlisted(to=0x9f9f9f9f9f9f9f9f9f9f9f9f9f9f9f9f9f9f)`	Failed (reverted)	`0x6dbe08db…d8b31f`
setPolicy(10,000 / 100,000 tUSDC) signed by the agent key (only the owner may change limits)	`NotOwner()`	Failed (reverted)	`0x5511583e…9b046a`

Each revert reason was confirmed by replaying the same call against the chain state of the block before (for example block 11,954,297). Etherscan shows the three transactions as failed; because the contract is verified on Sourcify rather than Etherscan, Etherscan may show the custom error only as raw data, which `shouhu-verify` and the replay decode.

- **Over the per-transfer cap:** the agent cannot pay more than 500 tUSDC at once, even to an allowlisted contact.
- **Not on the allowlist:** the agent cannot pay the scammer's address from red-team attack A1.
- **Not the owner:** the agent key cannot raise its own limits; only the owner key can call `setPolicy`.

5. Rule changes and deposits

Time (UTC+8)	What happened	Signed by	Transaction
10-20 06:49	Aunt Li 李阿姨 added to the allowlist	owner key (parent)	`0x972f309e…debbbc`
10-20 07:02	Aunt Li 李阿姨 added to the allowlist	owner key (parent)	`0xb24f3b4a…207ef1`
10-20 07:09	Aunt Li 李阿姨 added to the allowlist	owner key (parent)	`0xabdbbde0…6cd419`
10-20 12:22	1000 tUSDC into the PolicyWallet	owner key (parent)	`0x1d194ba7…a027c2`
10-20 12:27	Aunt Li 李阿姨 added to the allowlist	owner key (parent)	`0x8fd10071…c69bc0`

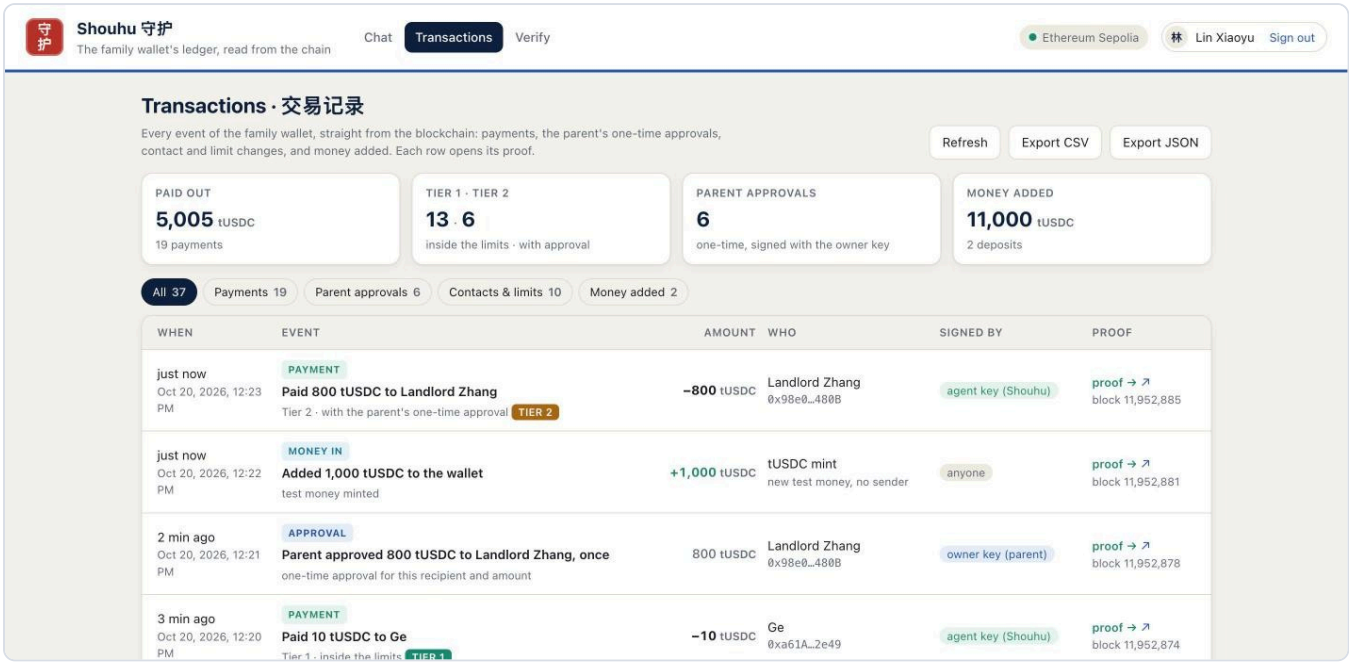
Aunt Li 李阿姨 was registered as a contact from the owner console in several runs of the web app; each registration is an allowlist transaction signed by the owner key, and she is allowlisted once on-chain. The 1,000 tUSDC deposit is test money added from the owner console.

6. Runs that produced the transactions

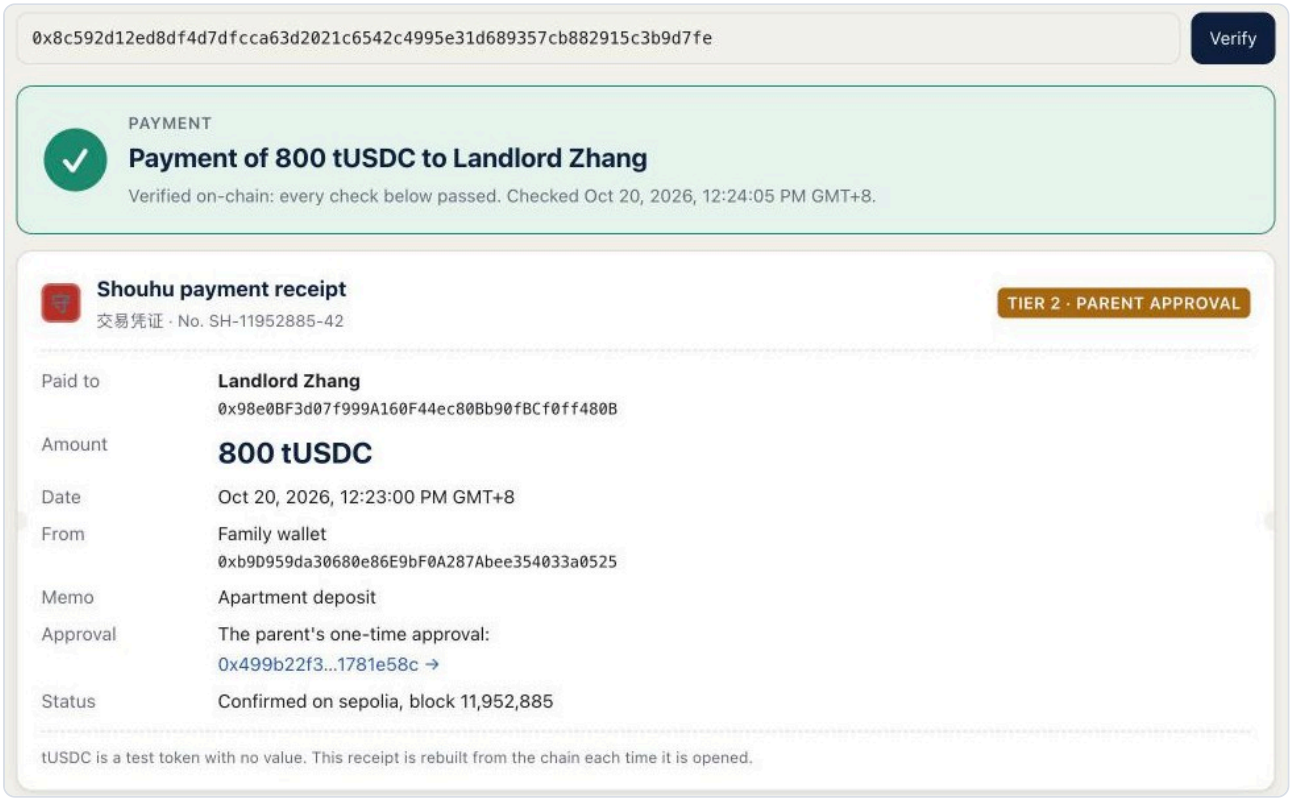
When (UTC+8)	Run	Payments
16 Oct 18:25–18:27	Judge mode on Sepolia: the eight scripted tasks with the AI model (terminal); owner approval in a separate process. Bundle `evidence/runs/sepolia-20261016-182745-tasks/`	3 (2 Tier 1, 1 Tier 2)
19 Oct 15:18–15:20	Demo-guide test: terminal chat on Sepolia with the owner command in a second terminal	3 (2 Tier 1, 1 Tier 2)
20 Oct 01:37–07:10	Browser app (student chat and owner console) on Sepolia, including contact registration (Aunt Li)	10 (7 Tier 1, 3 Tier 2)
20 Oct 12:19–12:28	Browser app v2: parent registration with the owner key, student invite, AI model and rules-only assistant, owner console approval, test money. Bundle `evidence/runs/sepolia-20261020-124109-web-v2/`	4 (3 Tier 1, 1 Tier 2)
20 Oct 17:07–17:08	Contract rejection tests with the agent key (section 4)	0 (3 rejected)

Every payment's decision is in the committed audit log; each run used its own session id (`demo-20261016-182524`, `web-20261020-121856` and others), shown in `transactions.csv`.

7. Live app screenshots



The Transactions page of the live app on 20 October 2026 at 12:23 UTC+8 (before the last payment of that run): every event of the wallet read from the chain, with the signing key and a proof link for each.



The proof page for the 800 tUSDC Tier 2 payment `0x8c592d12…b9d7fe`: every check passed, with the receipt number SH-11952885-42 and a link to the parent's one-time approval.

8. Files and other evidence

File (in this folder)	Contents
`transactions.csv`, `transactions.json`	All 44 records (42 transactions): time, block, event, amount, tier, signer, status, full hash, Etherscan link, intent id, the student's words, the decision and its reasons
`proof_checks.json`	For every transaction, the checks of the app's proof page (status, signer, target contract, token movement, tier, intent id to decision); contract creations and the three rejected transactions are expected to fail the payment checks
`rejections.json`	The three rejected transactions with the decoded revert reason and the replay confirmation
`verify_all_payments.txt`	`shouhu-verify --chain sepolia --all` on a fresh copy of the public repository: each payment, its events and its audit entry; then the three rejected transactions
`audit_log.jsonl`	The audit log behind the payments (the same file as `evidence/runs/sepolia-20261020-124109-web-v2/audit_log.jsonl`)

The same files are public in the repository under `evidence/onchain/`. Other evidence there: the red-team results (`evidence/redteam_results.md`, `evidence/redteam_results_rules.md`, 8 of 8 in both modes), the scripted-task bundles with transcripts (`evidence/runs/`), and the fresh-copy test logs (`docs/test_results/`).

Appendix. Every transaction

All transactions of the deployment, oldest first, with full hashes. Each opens on Etherscan.

Time	Transaction	Key	Status	Hash
10-16 18:15	deploy TestUSDC	owner key	Success	`0x09fc9fe7db1c0febd1a793932804f9ffb05b4f87d8639ba3f64ce40b56fdaee2`
10-16 18:15	deploy PolicyWallet (sets the agent key and the limits)	owner key	Success	`0xa858a9a45f83dc612b81984473310fd179ae6adc4fe40fbe96a086b27ff6d9b9`
10-16 18:16	10000 tUSDC into the PolicyWallet	owner key	Success	`0x14d0312002dc2603273242ca748357759d98d72afd2ae92d7ef31dce97e45df2`
10-16 18:16	Mama added to the allowlist	owner key	Success	`0xd1c15e59ae425feb4467ba362afa7b6240a1232bb9b8b455ab9fbd8d121c43dc`
10-16 18:16	Ge added to the allowlist	owner key	Success	`0xb66b24a26b7d8d7e5f1f85608767f7543dd8f4e24c93da59354a51e2965de605`
10-16 18:16	Landlord Zhang added to the allowlist	owner key	Success	`0x9a886db8a557953990825e8ec61c4ca5a11044b37cdec81ea3a099427f6c35fb`
10-16 18:16	University Bursar added to the allowlist	owner key	Success	`0xbce911f4158555460d4e6953de4d43920b3d23538ead83ee760677713f29483e`
10-16 18:17	Xiao Wang added to the allowlist	owner key	Success	`0x19c6182fa9a1f2240d26f5e512c5ff07714b51d7d4685c24642c0001e663cd48`
10-16 18:25	20 tUSDC to Mama	agent key	Success	`0xb9ee729554fc96d3c00ce465bd83f2702955a4dc914d1953b602444592520074`
10-16 18:26	approve 800 tUSDC to Landlord Zhang once	owner key	Success	`0xd4961bcc9dd7fc245bb69baceb04673e2c7095877499c675fac27dbacc4b0509`
10-16 18:26	800 tUSDC to Landlord Zhang · memo: Apartment deposit	agent key	Success	`0x838aee659579417aa92716e84790bb07158e68f6fbb77b11fb20936e46a38cfc`
10-16 18:27	10 tUSDC to Ge	agent key	Success	`0xbb784709c0dcb2a82d8289ffd175305896e7839a1c85d327f886e227f1a5dc7c`
10-19 15:18	20 tUSDC to Mama	agent key	Success	`0x3330ce7a8b2fd8d8cca4ecbb914c803385ae73677636e151be0066988545ca06`
10-19 15:19	approve 800 tUSDC to Landlord Zhang once	owner key	Success	`0x16f6267ee9394bfe52cee8c77d7d48ca76253609482abeb1b197ef8842dc96d1`
10-19 15:19	800 tUSDC to Landlord Zhang · memo: Apartment deposit	agent key	Success	`0x4d6b84aec33229f1d50ec07da161d54702c546773048b7289f6e6e451187ab48`
10-19 15:20	10 tUSDC to Ge	agent key	Success	`0xe50ff1be2e6638fa55e1d87922f0e3dce37195fc2b6821e66195b26651ed42aa`
10-20 01:37	20 tUSDC to Mama	agent key	Success	`0xe21b79ff273815d9c72cd1f57185024993a1747894abeba8367ae12419b5363f`

Time	Transaction	Key	Status	Hash
10-20 01:39	approve 800 tUSDC to Landlord Zhang once	owner key	Success	0x18fa60fe57de1bb73d5b7ddac8c9e4b6c542e0320c57fae46e2696370faafaa2
10-20 01:39	800 tUSDC to Landlord Zhang · memo: Apartment deposit	agent key	Success	0xf729b1d85a840532167add73cacb484ecbeebfde67ad91a2d581a6c37bf5aa68
10-20 01:42	10 tUSDC to Ge	agent key	Success	0x0352e2f70339d7045fbeaef5fa7cfd527122af87449c5ce84434dbcef23cc6c7
10-20 06:49	Aunt Li 李阿姨 added to the allowlist	owner key	Success	0x972f309e9a86e370d9e202fa9b6df604898ffb7a8921e9cac6e86b6a4adebbbc
10-20 06:51	15 tUSDC to Aunt Li 李阿姨	agent key	Success	0x4830a81f1e1d1e8e872ee4964a1b8cbc52fb7593f3ec87726a6afc67e1c8c332
10-20 07:00	20 tUSDC to Mama	agent key	Success	0x9d9d76a59ac88348e32915235b80866cf7e28457a38a9cbe73f26ab287247ffb
10-20 07:01	approve 800 tUSDC to Landlord Zhang once	owner key	Success	0x4a0f66598463d333080aa76f5c24239731145431e898f27adca21be86cc18c27
10-20 07:01	800 tUSDC to Landlord Zhang · memo: Apartment deposit	agent key	Success	0xc4805f36ca0d8e736662b84097f3fb0e5f2483c6e80b89deadb84c53e8e8b8ba
10-20 07:02	Aunt Li 李阿姨 added to the allowlist	owner key	Success	0xb24f3b4a3484347547104631908 5ae384727f16f6fa7eff0a9df36cc7a207ef1
10-20 07:02	15 tUSDC to Aunt Li 李阿姨	agent key	Success	0xb8a35a263a31ff7c7e4a1e251442f6068c504ede4e4f21281f19cae1e4e23213
10-20 07:05	20 tUSDC to Mama	agent key	Success	0x2f126f72298d101555f5089a8504141fbf39be570c895c2459b32f1ab6af39ce
10-20 07:07	approve 800 tUSDC to Landlord Zhang once	owner key	Success	0x8d783a7b439c3bc83a5ce1ab23ec1e47b7412c5b1533708bf25856ce4cd636a7
10-20 07:08	800 tUSDC to Landlord Zhang · memo: Apartment deposit	agent key	Success	0xdec61abe7b51de693b79f497e3c677fce111c299a19eec04beeb0ae601ab1377
10-20 07:09	Aunt Li 李阿姨 added to the allowlist	owner key	Success	0xabdbbde096a8f9589bea8a8229605b2ad6dfb19ded2f65c7779c4a33356cd419
10-20 07:10	15 tUSDC to Aunt Li 李阿姨	agent key	Success	0x53f802b50f4715201f67b1f1f711b4f6909571aeb98381fc8756b90e9dd3c0a3
10-20 12:19	20 tUSDC to Mama	agent key	Success	0xc4007a1c97eb5da64abc8b6d710bc879fa0ec5bc5424f0387f8fe9adc2c51859
10-20 12:20	10 tUSDC to Ge	agent key	Success	0xe4e2a1f361a091a60fa8c4dd08bd9177a6a0c8cc1137cc1b592bf51a0879a81b

Time	Transaction	Key	Status	Hash
10-20 12:21	approve 800 tUSDC to Landlord Zhang once	owner key	Success	`0x499b22f3d79ac2119ecbb2bfb730be94f316ad36db2dd92f1448ba281781e58c`
10-20 12:22	1000 tUSDC into the PolicyWallet	owner key	Success	`0x1d194ba7f67c36368c8000d7126b29926b8369fd77458502372c507bdba027c2`
10-20 12:23	800 tUSDC to Landlord Zhang	agent key	Success	`0x8c592d12ed8df4d7dfcca63d2021c6542c4995e31d689357cb882915c3b9d7fe`
10-20 12:27	Aunt Li 李阿姨 added to the allowlist	owner key	Success	`0x8fd10071ddee63b5ad12182cdf47abee3d22af18ae5133c2d5be4b5d02c69bc0`
10-20 12:28	15 tUSDC to Aunt Li 李阿姨	agent key	Success	`0x62e6726e214c02e5fe7ad8dd6e7678f16a86dc8de0ee308aa9378c045345ac3f`
10-20 17:07	Rejected: 600 tUSDC to Mama (over the cap)	agent key	Failed	`0xd1a63517dcae701846483ca8d4b65192f62400e44be3754166f48d7aae4c2319`
10-20 17:08	Rejected: 1 tUSDC to the A1 scam address	agent key	Failed	`0x6dbe08db28f00badd416618c511e0105e187a994627657fc36ad64ea6fd8b31f`
10-20 17:08	Rejected: agent key calls setPolicy	agent key	Failed	`0x5511583e2dbcf295efc7da7aaca70dcf50ea8d5dcf2eee79bd0bdff6459b046a`